

Nathan Wakefield

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Recent Computer Science graduate with experience in embedded systems, automation, and cross-platform integration. Skilled in C++, Python, and real-time systems development. Seeking opportunities that bridge software and electrical engineering.

Education

California State University, Long Beach
B.S. Computer Science

Aug. 2019 – Dec. 2024

Relevant Coursework

Databases • Algorithms • Data Structures • Object-Oriented Design • Python • Software Design Principles • Operating Systems • Intro and Advanced Computer Security • Networks and Network Security

Projects

Lead Developer | SAE Baja Data Collection System | CSULB April 2024 - May 2025

- Developed a data acquisition system using ESP32s, ESPNow, and C/C++ to collect real-time vehicle performance data from sensors like GPS and suspension height.
- Integrated direct pin wiring and I2C communication between sensors, with data logged to an SD card for post-processing and visualization in Blender.
- Developed LoRa communication to a Raspberry Pi for real-time monitoring and future UI implementation.

Designer & Developer | Digital Quadrascopic Lenticular Camera | Self-Initiated Dec. 2023

- Developed software in Python to control the camera, handle photo processing, and manage I/O operations.
- Integrated a Raspberry Pi and specialized HAT to synchronize and control four cameras simultaneously.
- Designed and 3D printed a custom body using TinkerCAD and Fusion 360.

Developer | Customer Review Request Automation | DeLillo Chevrolet Feb 2023

- Automated customer review requests, saving 4 hours weekly by parsing CSVs and automating webpage interaction.
- Utilized NumPy for data preprocessing and CSV manipulation, optimizing workflow and reducing errors.

Work Experience

IT Manager / Software Developer | Sowing Seeds Health | Irvine, CA Jan. 2025 - Present

- Oversaw the installation, configuration, and maintenance of hardware and software systems across multiple locations.
- Provided daily IT support for 40+ employees, as well as trained staff on system use.
- Rebuilt and managed the organization's website using WordPress and Beaver Builder
- Supporting data collection for Enhanced Care Management and Community Health Worker programs.

3D Printing Technician | Self-Employed | Long Beach, CA Jan. 2019 - Present

- Facilitated the process of printing and shipping prototypes to professionals around the US.
- Constructed CAD models using TinkerCAD and Fusion 360 and produced them using PLA, ABS, and resin
- Readily maintained a group of 3D printers, tasks including frequent bed leveling and build plate cleaning.
- Configured open-source tooling to automate internet-driven print jobs and optimize workflow.

Marketing Manager | DeLillo Chevrolet | Huntington Beach, CA Jan. 2021 - May 2024

- Performed data analysis from large datasets of customer data using Excel's visualization stack.
- Assisted with maintenance and upkeep of backend price management software.
- Delivered in-depth business marketing initiatives to drive sales and generate leads.

Leadership and Activities

Data Acquisition Lead | SAE Baja Team | CSULB April 2024 – Present

- Led a team of 5+ CS, CE, and EE students in developing an embedded system for Baja vehicle data collection.
- Mentored the team in sensor integration, wireless and wired communication, and programming for embedded systems.
- Completed the first iteration of the data collection and information relay system for SAE Baja in Arizona.

Technical Skills

Software: Microsoft Office Suite, Google Suite, GitHub, Fusion360, Solidworks, X3D, KiCAD, Arduino IDE

Languages: Python, JavaScript, C/C++

Tools: Arduino IDE, VS Code, Visual Studio, Node.js, Tailwind; **OS:** Windows 10/11, Ubuntu, MacOS Kali Linux, Linux Mint, Linux Server